



Joint Parameter Estimation in the Pareto Model under Type-II Censoring: A Bayesian Approach

Akbar Asgharzadeh¹, Ali Saadati Nik²

^{1,2}Department of Statistics, Faculty of Mathematical Sciences, University of Mazandaran, Babolsar,
Iran

Abstract:

This paper presents a Bayesian approach for joint estimation of the two-parameter Pareto model under type-II censoring. Based on censored lifetime data, the posterior distributions of the model parameters are derived by combining the likelihood function with suitable prior distributions. A two-step procedure is developed to obtain closed-form Bayesian credible regions for the parameters.

Keywords: Pareto distribution, Bayesian credible regions, type-II censored data.

Mathematics Subject Classification (2010): 99X99, 99X99, 99X99.

1 Introduction

Let us consider a two-parameter Pareto distribution with the shape parameter α and scale parameter σ , denoted by $\mathcal{P}(\alpha, \sigma)$, with the cumulative distribution function (cdf) and the probability density function (pdf)

$$\mathcal{F}(x; \alpha, \sigma) = 1 - (\sigma x)^{-\alpha}, \quad x \geq \frac{1}{\sigma}, \quad \alpha > 0, \quad (1.1)$$

$$f(x; \alpha, \sigma) = \alpha \sigma (\sigma x)^{-\alpha-1}, \quad x \geq \frac{1}{\sigma}, \quad \alpha > 0, \quad (1.2)$$

¹Corresponding Author: a.asgharzadeh@umz.ac.ir

respectively.

In a life-testing experiment, suppose that n units are placed under observation and the test is terminated upon the occurrence of the m -th failure, where $m < n$. Consequently, only the first m failure times are observed. This ordered sample, denoted as $X_{1:n} < X_{2:n} < \cdots < X_{m:n}$, is termed a type-II censored sample. For ease of notation, we represent $X_{1:n}, \dots, X_{m:n}$ as $X_1, \dots, X_m$.

In this paper, Bayesian credible regions for the parameters of the two-parameter Pareto distribution are developed based on type-II censored data.

Confidence intervals and confidence regions provide plausible ranges for unknown parameters and are widely used across various disciplines. For example, in quality control and industrial applications, they play a crucial role in assessing product lifetimes and determining appropriate warranty periods. Numerous authors have studied the construction of confidence intervals and regions for the unknown parameters; see, for example, [Asgharzadeh et al. \(2017, 2020\)](#), [Fernández \(2012, 2013, 2023\)](#), [Lennartz et al. \(2021\)](#), [Saadati Nik and Asgharzadeh \(2025\)](#); [Saadati Nik et al. \(2025\)](#), and [Wu \(2022\)](#).

Suppose that $\mathbf{X} = (X_1, \dots, X_m)$ is a type-II censored sample from the $\mathcal{P}(\alpha, \sigma)$ distribution with the observed sample $\mathbf{x} = (x_1, \dots, x_m)$. The joint pdf of $\mathbf{X} = (X_1, \dots, X_m)$ is

$$\begin{aligned} f(\mathbf{x} \mid \alpha, \sigma) &\propto \prod_{i=1}^m f(x_i; \alpha, \sigma) [1 - \mathcal{F}(x_m; \alpha, \sigma)]^{n-m} \\ &= \frac{1}{\prod_{i=1}^m x_i} \alpha^m \sigma^{-n\alpha} \exp(-\alpha(t_m + n \ln x_1)), \quad \sigma \geq \frac{1}{x_1}, \alpha > 0, \end{aligned} \quad (1.3)$$

where, t_m is the observed value of $T_m = \sum_{i=1}^m \ln X_i + (n - m) \ln X_m - n \ln X_1$. Clearly (T_m, X_1) is a minimal sufficient statistics for (α, σ) . For $m \geq 2$, the maximum likelihood (ML) estimates of α and σ , denoted by $\hat{\alpha}$ and $\hat{\sigma}$, are given by

$$\hat{\alpha} = \frac{m}{t_m}, \quad \text{and} \quad \hat{\sigma} = \frac{1}{x_1}.$$

It can be seen that the distribution of T_m is gamma $\mathcal{G}(m - 1, \alpha)$ distribution and $Y = \ln(\sigma X_1) \sim \text{Exp}(n\alpha)$ with $E(Y) = (n\alpha)^{-1}$.

Consider the joint prior distribution for parameters (α, σ) defined as:

$$\pi(\alpha, \sigma) \propto \alpha^c \sigma^{\gamma\alpha-1} b^{-\gamma\alpha} \exp(-d\alpha), \quad \alpha > 0, \quad \sigma \geq \frac{1}{b},$$

where γ, b, c , and d are positive hyperparameters. This prior is applied in Bayesian analysis of the two-parameter Pareto distribution, as noted in works by [Arnold and Press \(1989\)](#) and [Lwin \(1972\)](#). The joint prior can be factored as:

$$\pi(\alpha, \sigma) = \pi(\alpha) \pi(\sigma \mid \alpha),$$

with α following a gamma $\mathcal{G}(c, d)$, and $\sigma \mid \alpha \sim \mathcal{P}(\gamma\alpha, b)$.

By setting $\gamma = d = 0$, $c = -1$, and taking the limit as $b \rightarrow \infty$ ($b = x_1$ is sufficient), the prior simplifies to the non-informative form:

$$\pi(\alpha, \sigma) \propto \frac{1}{\alpha\sigma}, \quad \alpha > 0, \quad \sigma > 0.$$

So, the posterior pdf of (α, σ) given $\mathbf{X} = \mathbf{x}$ is defined by

$$\begin{aligned} \pi(\alpha, \sigma | \mathbf{x}) &= \frac{1}{c(\mathbf{x})} f(\mathbf{x} | \alpha, \sigma) \pi(\alpha, \sigma) \\ &= \frac{1}{c(\mathbf{x})} \alpha^{m+c} \sigma^{-\alpha(n+\gamma)-1} \exp(-\alpha(d + \gamma \ln b + t_m + n \ln x_1)), \end{aligned} \quad (1.4)$$

for $\sigma > \xi$, $\alpha > 0$, where, $\xi = \max(\frac{1}{x_1}, \frac{1}{b})$ and

$$c(\mathbf{x}) = \int_0^\infty \int_\xi^\infty f(\mathbf{x} | \alpha, \sigma) \pi(\alpha, \sigma) d\sigma d\alpha = \frac{\Gamma(m+c)}{(n+\gamma)(\omega + t_m + n \ln x_1)^{m+c}},$$

with, $\omega = d + \gamma \ln(b\xi) + n \ln \xi$.

Corollary 1.1. *The conditional posterior density of α given the data $\mathbf{x}$ is expressed as:*

$$\begin{aligned} \pi(\alpha | \mathbf{x}) &= \int_\xi^\infty \pi(\alpha, \sigma | \mathbf{x}) d\sigma \\ &= \frac{(\omega + t_m + n \ln x_1)^{m+c}}{\Gamma(m+c)} \alpha^{m+c-1} \exp(-\alpha(\omega + t_m + n \ln x_1)), \end{aligned}$$

for $\alpha > 0$. Evidently, $\alpha | \mathbf{x}$ follows a gamma $\mathcal{G}(m+c, \omega + t_m + n \ln x_1)$.

Corollary 1.2. *Using the posterior distribution, the conditional posterior density of σ given α and the data $\mathbf{x}$ is derived as:*

$$\pi(\sigma | \alpha, \mathbf{x}) = \frac{\pi(\alpha, \sigma | \mathbf{x})}{\pi(\alpha | \mathbf{x})} = \alpha(n+\gamma) \xi^{\alpha(n+\gamma)} \sigma^{-\alpha(n+\gamma)-1}, \quad \sigma > \xi.$$

It follows that the transformed variable $Y = \frac{\xi}{\sigma} | (\alpha, \mathbf{x})$ follows a Beta distribution, specifically $\text{Beta}(\alpha(n+\gamma), 1)$.

2 Credible regions

This section outlines a two-step procedure to construct a $100(1 - \zeta)\%$ Bayesian credible region for the parameters (α, σ) , leveraging Corollaries 1.1 and 1.2. The algorithm proceeds as follows:

Step 1. Determine a $100(\sqrt{1 - \zeta})\%$ credible interval Λ for α using the marginal posterior distribution $\pi(\alpha | \mathbf{x})$ from Corollary 1.1. Given that $\alpha | \mathbf{x} \sim \mathcal{G}(m+c, \omega + t_m + n \ln x_1)$, the transformed variable $2(\omega + t_m + n \ln x_1)\alpha | \mathbf{x}$ follows a chi-square distribution with $2m + 2c$ degrees of freedom, i.e., $2(\omega + t_m + n \ln x_1)\alpha | \mathbf{x} \sim \chi^2(2m + 2c)$. We have

$$\Pr \left[\chi_{\zeta-}^2(2m + 2c) < 2(\omega + t_m + n \ln x_1)\alpha < \chi_{\zeta+}^2(2m + 2c) | \mathbf{x} \right] = \sqrt{1 - \zeta},$$

which is equivalent to:

$$\Pr \left[\frac{\chi_{\zeta^-}^2 (2m + 2c)}{2(\omega + t_m + n \ln x_1)} < \alpha < \frac{\chi_{\zeta^+}^2 (2m + 2c)}{2(\omega + t_m + n \ln x_1)} \mid \mathbf{x} \right] = \sqrt{1 - \zeta},$$

where, $\zeta^- = \frac{1 - \sqrt{1 - \zeta}}{2}$ and $\zeta^+ = \frac{1 + \sqrt{1 - \zeta}}{2}$. Thus, the credible interval for α is:

$$\Lambda = \left\{ \alpha : \frac{\chi_{\zeta^-}^2 (2m + 2c)}{2(\omega + t_m + n \ln x_1)} < \alpha < \frac{\chi_{\zeta^+}^2 (2m + 2c)}{2(\omega + t_m + n \ln x_1)} \mid \mathbf{x} \right\}.$$

Step 2. For each $\alpha \in \Lambda$, construct a $100(\sqrt{1 - \zeta})\%$ credible interval $\mathcal{D}(\alpha)$ for σ based on the conditional posterior $\pi(\sigma \mid \alpha, \mathbf{x})$ from Corollary 1.2. Since $\pi(\sigma \mid \alpha, \mathbf{x})$ is an decreasing function of σ over the interval (ξ, ∞) , define the interval as:

$$\mathcal{D}(\alpha) = \left\{ \sigma \in (\xi, k) : \int_{\xi}^k \pi(\sigma \mid \alpha, \mathbf{x}) d\sigma = \sqrt{1 - \zeta} \right\},$$

yielding the solution $k = \xi / (1 - \sqrt{1 - \zeta})^{\frac{1}{\alpha(n+\gamma)}}$.

The $100(1 - \zeta)\%$ Bayesian credible region for (α, σ) , denoted $\mathcal{C}^{\text{Bys}}$, is given by:

$$\mathcal{C}^{\text{Bys}} = \left\{ (\alpha, \sigma) : \frac{\chi_{\zeta^-}^2 (2m + 2c)}{2(\omega + t_m + n \ln x_1)} < \alpha < \frac{\chi_{\zeta^+}^2 (2m + 2c)}{2(\omega + t_m + n \ln x_1)}, \right. \\ \left. \xi < \sigma < \frac{\xi}{(1 - \sqrt{1 - \zeta})^{\frac{1}{\alpha(n+\gamma)}}} \right\}. \quad (2.1)$$

The area of the region $\mathcal{C}^{\text{Bys}}$ is

$$\text{Area}[\mathcal{C}^{\text{Bys}}] = \iint_{\mathcal{C}^{\text{Bys}}} d\sigma d\alpha \\ = \xi \int_{\frac{\chi_{\zeta^-}^2 (2m + 2c)}{2(\omega + t_m + n \ln x_1)}}^{\frac{\chi_{\zeta^+}^2 (2m + 2c)}{2(\omega + t_m + n \ln x_1)}} \left((1 - \sqrt{1 - \zeta})^{\frac{-1}{\alpha(n+\gamma)}} - 1 \right) d\alpha. \quad (2.2)$$

3 Illustrative example

In this section, an numerical example is provided for illustration. The example is based on a Pareto data set reported by [Hong et al. \(2008\)](#) on business failures. The underlying random variable, X , denotes the length of time in years for which a business operates until failure. Only the 65 smallest failure times were observed in a sample of 200 businesses. Business operational lifetimes are displayed in Table 1. For this data, $x_1 = 1.02$ is the first failure time and $t_m = 654.264$ is the observed value of T_m .

To derive the Bayesian credible regions, we consider the non-informative prior. Assuming that the confidence level $1 - \zeta$ is 0.90 and 0.95, the corresponding 95% and 90% Bayesian credible regions $\mathcal{C}^{\text{Bys}}$ are

Table 1: Business operational lifetimes

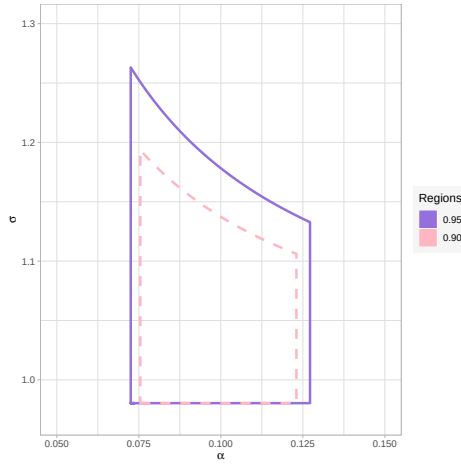
1.02	1.08	1.09	1.11	1.12	1.22	1.42	1.45	1.57	1.59	1.62	1.62	1.74
1.94	1.94	1.96	1.99	2.08	2.32	2.88	3.01	3.22	3.65	3.75	4.11	4.15
4.26	4.38	4.71	4.80	5.47	5.74	6.38	6.74	7.42	7.44	7.60	7.81	8.46
8.65	8.92	10.93	13.02	15.19	16.16	16.23	19.59	19.81	20.16	20.64	21.32	23.67
24.04	25.41	28.77	30.20	30.66	31.29	34.37	45.04	47.62	48.14	50.74	51.50	52.68

obtained as follows:

$$\mathcal{C}_{0.95}^{\text{Bys}} = \left\{ (\alpha, \sigma) : 0.0725 < \alpha < 0.1271, 0.9803 < \sigma < \frac{0.9803}{(0.0253)^{\frac{1}{200\alpha}}} \right\},$$

$$\mathcal{C}_{0.90}^{\text{Bys}} = \left\{ (\alpha, \sigma) : 0.0754 < \alpha < 0.1230, 0.9803 < \sigma < \frac{0.9803}{(0.0513)^{\frac{1}{200\alpha}}} \right\}.$$

The areas of $\mathcal{C}_{0.95}^{\text{Bys}}$ and $\mathcal{C}_{0.90}^{\text{Bys}}$ are 0.0111 and 0.0077, respectively. For comparison, the 95% and 90% Bayesian regions $\mathcal{C}^{\text{Bys}}$ are illustrated in Figure 1.

Figure 1: Bayesian confidence regions for (α, σ) .

Acknowledgments

This research was financed by a research grant from the University of Mazandaran.

References

- Arnold, B.C. and Press, S.J. (1989). Bayesian estimation and prediction for Pareto data, *Journal of the American Statistical Association*, 84, 1079-1084.
- Asgharzadeh, A., Fernández, A.J., & Abdi, M. (2017). Confidence sets for the two-parameter Rayleigh distribution under progressive censoring. *Applied Mathematical Modeling*, 47, 656-667.
- Asgharzadeh, A., Bagheri, S.F., Ibrahim, N.A., & Abubakar, M.R. (2020). Optimal confidence regions for the two-parameter exponential distribution based on records. *Computational Statistics*, 35, 309–326.
- Fernández, A.J. (2012). Minimizing the area of a Pareto confidence region. *European Journal of Operational Research*, 221, 205-212.
- Fernández, A.J. (2013). Smallest Pareto confidence regions and applications. *Computational Statistics & Data Analysis*, 62, 11–25.
- Fernández, A.J. (2023). Optimal Confidence Regions for Weibull Parameters and Quantiles under Progressive Censoring. *Algorithms*, 16, 427. <https://doi.org/10.3390/a16090427>
- Hong, C.-W., Wu, J.-W., Cheng, C.-H. (2008). Computational procedure of performance assessment of lifetime index of Pareto lifetime businesses based on confidence interval. *Applied Soft Computing*, 8, 698–705.
- Lennartz, J.M., Bedbur, S., and Kamps, U. (2021). Minimum area confidence regions and their coverage probabilities for type-II censored exponential data. *Statistical Papers*, 62, 171–191. <https://doi.org/10.1007/s00362-019-01087-x>
- Lwin, T. (1972). Estimation of the tail of the Paretian law, *Skand Aktuarietidskr*, 55, 170-178.
- Saadati Nik, A., and Asgharzadeh, A. (2025). Bayesian joint estimation for the two-parameter exponential distribution based on records. *Communications in Statistics – Theory and Methods*, 1–15. <https://doi.org/10.1080/03610926.2025.2560035>
- Saadati Nik, A., Asgharzadeh, A., and Fernández, A.J. (2025). Bayesian credible regions for two-parameter exponential distributions under Type-II censoring. *Journal of Computational and Applied Mathematics*, 470, 116721. <https://doi.org/10.1016/j.cam.2025.116721>

Wu, S.F. (2022). Interval estimation for the two-parameter exponential distribution under progressive type-II censoring on the Bayesian approach. *Symmetry*, 14, 808. <https://doi.org/10.3390/sym14040808>